CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Third Semester of B.Sc.(Physics) Theory Examination March 2019
PD 201 Mathematical Physics-II

Date: 14/03/2019,Thursday **Time**: 01:30 p.m. to 02:00 p.m. **Maximum Marks**: 20

MCQ

Important Instructions:

- Provide all necessary information as required in the answer sheet.
- Attempt all questions.
- Answer the questions in the question paper itself.
- Use of cell phones is strictly prohibited.
- Use of non-programmable calculators may be allowed if needed.

Q.1. Choose the correct answer for the following questions. [20]

1. Which of the following possible valid series solution of the differential equation $(x-3)(x-7)(x-5)y'' - 3xy' + 5y = 0$?

   a. $\sum_{n=0}^{\infty} a_n x^n$

   b. $\sum_{n=0}^{\infty} a_n (x-7)^n$

   c. $\sum_{n=0}^{\infty} a_n (x-5)^n$

   d. $\sum_{n=0}^{\infty} a_n (x-3)^n$

2. Which of the following point is not an ordinary point of the differential equation $(x-3)y'' - (x-1)(x-2)y' + xy = 0$?

   a. 0 b. 1 c. 2 d. 3

3. For Legendre polynomial $P_n(x)$, $\int_{-1}^{1} [P_7(x)]^2\, dx =$__________.

   a. $\frac{2}{7}$ b. $\frac{2}{15}$ c. 1 d. 0

4. For Legendre polynomial $P_n(x)$, $P_0(7) =$__________.

   a. 0 b. 1 c. 2 d. 7

5. For Bessel function $J_n(-x) =$__________, n is odd.

   a. 0

   b. $J_n(x)$

   c. $-J_n(x)$

   d. not defined

6. For Bessel function $J_n(x)$, $\lim_{x \to 0} \frac{J_1(x)}{x} =$__________.

   a. $\frac{1}{2}$ b. $\frac{1}{4}$ c. $\frac{1}{8}$ d. 1

7. Which of the following is two dimensional wave equation?

   a. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{\partial z}{\partial t}$

   b. $\frac{\partial^2 z}{\partial x^2} = \frac{\partial z}{\partial t}$

   c. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial t^2}$

   d. $\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial t^2}$

8. The equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$ is known as__________.

   a. two dimensional heat equation

   b. one dimensional wave equation

   c. two dimensional wave equation

   d. Laplace equation

9. For solving one dimensional heat equation $\frac{\partial z}{\partial t} = \frac{\partial^2 z}{\partial x^2}$, the number of initial and boundary conditions required are ________ respectively.

   a. 2 and 2 b. 2 and 1 c. 1 and 2 d. 1 and 3

10. Which of the following function is an even function?

   a. $\frac{x-\sin x}{x^2}$
   b. $\frac{x^4-\cos x}{x^6}$
   c. $x^3 \cos 2x$
   d. $\frac{x^3}{e^{x^2}}$

11. In Fourier series expansion of $f(x) = x^5$ in $[-7, 7]$,________.
   a. only sine terms are present
   b. only cosine terms are present
   c. only constant term is present
   d. both sine and cosine terms are present

12. The period of $\cos kt$ is________.

   a. $\frac{2\pi}{k}$ b. $2k$ c. $\frac{\pi}{k}$ d. 2π

13. $\int_{-\pi}^{\pi} x \cos x \, dx$=________.

   a. 1 b. 2 c. 3 d. 0

14. Which of the following error is caused by careless handling?

   a. Systematic error
   b. Gross error
   c. Random error
   d. none of these

15. If x is the true value of a number and $\bar{x}$ is the approximate value obtained by computation, then the relative error is ________.

   a. $|x - \bar{x}|$
   b. $|\frac{x-\bar{x}}{x}|$
   c. $|\frac{x-\bar{x}}{x}| \times 100$
   d. $|\frac{\bar{x}}{x}|$

16. For gamma function $\Gamma(\alpha)$, $\Gamma(\frac{1}{2})$________.

   a. π b. $\sqrt{\pi}$ c. 1 d. 0.

17. For gamma function $\Gamma(n)$, which of the following is not true?

   a. $\Gamma(n+1) = n\Gamma(n)$
   b. $\Gamma(n+1) = n!$
   c. $\Gamma(n+1) = (n+1)!$
   d. $\Gamma(-0.000001)$ is exist.

18. For beta function $B(m, n)$, $B(2,3)$=__________.

a. $\frac{1}{24}$ b. $\frac{1}{12}$ c. $\frac{1}{2}$ d. $\frac{1}{3}$

19. Which of the following is called complementary error function of x?

a. $\frac{2}{\pi}\int\limits_0^x e^{-t^2}\,dt$

b. $\frac{2}{\pi}\int\limits_x^\infty e^{-t^2}\,dt$

c. $\frac{2}{\sqrt{\pi}}\int\limits_0^x e^{-t^2}\,dt$

d. $\frac{2}{\sqrt{\pi}}\int\limits_x^\infty e^{-t^2}\,dt$

20. For an error function $erf(x)$, $erf(0)$=__________.

a. 0

b. 1

c. 2

d. not exist

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Date: 14/03/2019,Thursday **Time**: 02:00 p.m. to 4:30 p.m. **Maximum Marks**: 50

Instructions:

1. **Section I and II must be attempted in seperate answer sheets.**
2. **Make suitable assumptions and draw neat figures wherever required.**
3. **Use of non-programmable calculator is allowed.**
4. **Show necessary calculations.**
5. **Figures to right indicate marks.**

SECTION-I

Q-2 Answer the following questions as directed. [20]

(a) State a generating function of Legendre's polynomial $P_n(x)$. Prove that $P_n(1) = 1$. [03]

OR

(a) Write the series form of a Bessel's function $J_n(x)$. Show that $\lim\limits_{x\to 0} \frac{J_n(x)}{x^n} = \frac{1}{2^n\Gamma(n+1)}$; $(n > -1)$. [03]

(b) Evaluate the integral $\int\limits_0^\infty \sqrt[4]{x}e^{-\sqrt{x}}\, dx$. [02]

(c) Solve the equation $\frac{d^2y}{dx^2} - 100y = 0$ by the power series method. [03]

(d) The period of a simple pendulum is $T = 2\pi\sqrt{\frac{l}{g}}$, find the maximum error in T due to the possible error upto 1% in l and 2.5% in g. [02]

(e) Locate and classify the singular points on the x-axis for the differential equation $x(x-7)^5y'' - 2(x-7)y' - xy = 0$. [02]

OR

(e) Express the polynomial $5x^2 - 6x + 7$ in terms of Legendre polynomials. [02]

(f) Using the method of separation of variables, solve the partial differential equation $4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$, where $u(0, y) = e^{-5y}$. [04]

OR

(f) Using the method of separation of variables, solve the partial differential equation $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial y} + u$, where $u(x, 0) = 4e^{-3x}$. [04]

(g) Find the half range cosine series for the function $f(x) = x + 1$ for $0 < x < \pi$. [04]
Hence deduce that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \ldots = \frac{\pi^2}{8}$.

OR

(g) Find the Fourier series to represents the function $f(x)$ given by [04]

$$f(x) = \begin{cases} -11, \text{ for } -\pi < x < 0, \\ 11, \text{ for } 0 < x < \pi. \end{cases}$$

Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots = \frac{\pi}{4}$.

SECTION II

Q-3 Answer the following questions as directed. [30]

(a) (i) Evaluate the integral $\int_0^1 \frac{dx}{\sqrt{1-x^4}}$. [03]

(ii) For an error function $erf(x)$, show that $erf(\infty) = 1$. [02]

(b) By the method of least squares, find the straight line that best fits the following data: [05]

x	1	2	3	4	5
y	14	27	40	55	68

(c) For Legendre's polynomial $P_n(x)$, prove that [05]

(i) $(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$

(ii) $nP_n(x) = xP'_n(x) - P'_{n-1}(x)$.

OR

(c) For Bessel function $J_n(x)$, show that $J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$. [05]

(d) Find the series solution of differential equation $2x^2y'' + xy' - (x+1)y = 0$. [05]

OR

(d) Find the solution in series of differential equation $(1 - x^2)y'' - xy' + 4y = 0$ about $x = 0$. [05]

(e) The vibrations of an elastic string is governed by the partial differential equation $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$. The length of the string is π and the ends are fixed. The initial velocity is zero and the initial deflection is $u(x,0) = 2(\sin x + \sin 3x)$. Find the deflection $u(x,t)$ of the vibrating string for $t > 0$. [05]

OR

(e) Solve the partial differential equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with boundary condition $u(x,0) = 3\sin n\pi x, u(0,t) = u(l,t) = 0$, where $0 < x < l$. [05]

(f) Find the Fourier series expansion of the function defined in a single period by the relations $f(x) = \begin{cases} 1, & \text{for } 0 < x < \pi, \\ 2, & \text{for } \pi < x < 2\pi. \end{cases}$ [05]

OR

(f) Find the Fourier series expansion of the function $f(x) = x \sin x$ in $[-\pi, \pi]$. [05]

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